

Roll No. ....

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Total Pages : 3

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3453/NR

C-19/2115

**OBJECT ORIENTED PROGRAMMING  
USING C++**

**Paper-202**

**Sem.-III**

Time Allowed : 3 Hours]

[Maximum Marks : 50

**Note :** Attempt any **six** questions, selection three questions from Section A and B. Section-C is compulsory.

**SECTION-A**

1. How object-oriented language is different from structured programming? Explain.
2. Discuss various data types available in C++ with the help of suitable examples.
3. Define following control statements along with examples :
  - (a) For loop
  - (b) While loop
  - (c) Do loop.

3453/NR/162/W/710

[P. T. O.]

4. What is Function? Discuss its categories with the help of suitable examples.
5. How the security of object oriented systems implemented? Discuss.  $5 \times 3 = 15$

### SECTION—B

6. How Structure is different from any Array? Explain with the help of an example. How function can pass structure?

What is Function overloading? How is the function call matched with the function? Explain with the help of an example program.

8. What is a Pointer in C++? How it is declared? Is Pointer and Array related? Justify your answer.

Discuss different types of File organization. Describe various operations on files.

10. What is Polymorphism? Discuss its importance in object oriented programs.  $5 \times 3 = 15$

### SECTION—C

11. Define the following :

- (i) Object
- (ii) Inheritance
- (iii) Abstract class

- (iv) Union
- (v) Structure
- (vi) Function overloading
- (vii) Operator overloading
- (viii) Destructure
- (ix) Virtual function
- (x) IO streams.

$$10 \times 2 = 20$$

G-14/2051

SET-B

5260/MJ

**Object Oriented Programming using C++(CPE-103)**

**Instructions for the Candidate:** Section C is compulsory. Attempt any six questions selection three questions from each section A & B.

**Maximum marks: 50**

**Maximum Time: 3 hrs**

**Section-A**

- Q1) Explain all features of OOPS. (5)
- Q2) What are the types of Constructor and Explain them with examples? (5)
- Q3) Compare Procedural language and OOPS. (5)
- Q4) What do you understand by Operator Overloading .Explain with the help of an example?. (5)
- Q5) Explain nested class, Empty class and container class with their syntax. (5)

**Section-B**

- Q6) How we can pass Pointers to objects explain with the syntax and an example? (5)
- Q7) W.A.P to illustrate the use of File manipulation functions using file pointers (5)
- Q8) What is Inheritance and types of Inheritance and explain Ambiguity in Multiple inheritance . (5)
- Q9) Explain the concept of Function Overriding and Function Overloading with examples (5)
- Q10) What are Templates and explain their types with the help of an example. (5)

**Section-C (All carry 2 marks)**

Q1)

- a) What do u mean by NEW operator ?
- b) What is an Abstract Class and a Pure virtual Function?.
- c) What do you mean by this pointer?
- d) What is data hiding?
- e) What is a static object?.
- f) What is Exception Handling?
- g) What is a destructor?
- h) What is Type Conversion?

(Semester-II)

Time : Three Hours]

[Maximum Marks : 50

**Note :** Attempt any *six* questions by selecting *three* questions from each Section A and B. Section C is compulsory.

**SECTION-A**

- I. Explain the Concept of Function Overloading in C++ with suitable example.
- II. How memory management of Objects in a Class is performed? Explain in detail.
- III. Explain the working of Dynamic Constructor in C++ with suitable example.
- IV. Write a Programme in C++ for +Binary operator overloading.
- V. How Function Overriding is performed in C++? Explain in detail with suitable example. (3×5=15)

**SECTION-B**

- VI. Explain in detail the concept of :
  - (a) Array of Pointers.
  - (b) Dangling pointers and memory leak.

32510-MJ/500/HHH/443

W [P.T.O.]

- VIII. How files in C++ are accessed and updated? Explain in detail.
- IX. Explain the concept of Early and late binding with C++ Programs.
- X. Explain in detail the role of templates in C++. (3×5=15)

### **SECTION-C**

#### **(Compulsory Question)**

XI. Explain the following :

- (a) Hybrid Inheritance.
- (b) Static object.
- (c) Container class.
- (d) New and delete operators.
- (e) Friend class.
- (f) This pointer.
- (g) Pure virtual functions.
- (h) Destructor.
- (i) Scope resolution operator.
- (j) Pointer arithmetic.

(2×10=20)

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## SECTION-A

I. What are features of object oriented Programming? How is it better than its predecessor programming paradigms? Explain. 5

II. Explain the following :

(i) Function overloading.

(ii) Static data member. 5

5229-MJ/310/HHH/342

218 [P.T.O.]

III. Explain the concept of class and object. How do you create a class in C++? How do you access the members of a class? 5

IV. What are constructors and destructors? Explain different types of constructors with example. 5

 What do you understand by operator overloading? Write a program in C++ to overload '+' operator. 5

### SECTION-B

-  VI. What do you mean by inheritance? Explain different types of inheritance with the help of examples. 5
- VII. What is pointer? How a pointer is initialized? Which arithmetic operators can be used with pointers? Explain giving examples. 5
- VIII. What is the concept of binding in C++? Differentiate between early binding and late binding with the help of an example. 5
- IX. Discuss the concept of exception handling. Explain the use of try and catch for exception handling in C++. 5
- X. What are the various file streams available in C++? What are the various methods of opening files in C++? Explain giving examples. 5

## SECTION-C

### (Compulsory Question)

XI. Attempt the following questions :

- ~~(i)~~ What is an array?
- (ii) Distinguish between class and object.
- ~~(iii)~~ What do you mean by a container class?
- ~~(iv)~~ What is copy constructor?
- (v) How new and delete operators are used?
- ~~(vi)~~ Differentiate between private and protected access.
- (vii) What is type conversion?
- (viii) What is the purpose of this pointer?
- ~~(ix)~~ What is a pure virtual function?
- (x) What are function templates?

(10×2=20)